

Assignment 4

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Title:

Create any interactive data visualization dashboard for any real-life application like weather dashboard. Visualize data using charts and graphs (use libraries like Chart.js or [D3.js](#)).

Objectives:

- To design and develop an interactive stock market data visualization dashboard.
 - To visualize Sensex and Nifty index data using graphical charts and graphs.
 - To implement Chart.js library for dynamic and responsive data representation.
 - To provide weekly and monthly stock market views using interactive buttons.
 - To calculate and display summary statistics such as average price, highest value, and lowest value.
 - To enhance understanding of stock market trends through visual analysis.
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Outcomes:

- Developed an interactive web-based dashboard for stock market analysis.
- Visualized Sensex and Nifty data using line charts, bar charts, and doughnut charts.
- Gained practical knowledge of integrating Chart.js for real-time data visualization.
- Understood how dashboards help in analyzing financial market trends efficiently.
- Improved skills in HTML, CSS, and JavaScript for frontend development.
- Enhanced ability to interpret stock market data for better decision-making.

Theory:

Introduction to Stock Market Data Visualization:

Data Visualization is the graphical representation of financial and statistical data using charts, graphs, and dashboards. In the stock market, visualization plays a crucial role in analyzing price movements, trends, volatility, and overall market performance.

Stock market indices like **Sensex** and **Nifty** represent the performance of top companies listed on Indian stock exchanges. Since stock prices change frequently,

dashboards help investors and analysts monitor market trends effectively and make informed decisions.

Interactive dashboards allow users to:

- View real-time or historical stock data
 - Compare weekly and monthly performance
 - Analyze trends using graphical charts
 - Understand market movement patterns easily
-

About the Sensex and Nifty Dashboard

The Sensex and Nifty Interactive Data Visualization Dashboard is a web-based application developed using:

- **HTML** – To create the structure of the webpage
- **CSS** – To design and style the dashboard layout
- **JavaScript** – To implement logic, data handling, and interactivity
- **Chart.js** – To generate dynamic and responsive charts

The dashboard visually represents stock market data such as:

- Opening price
- Closing price
- High and Low values
- Daily index movement
- Weekly and Monthly trends

It also displays summary statistics like:

- Average closing price
 - Highest value in selected period
 - Lowest value in selected period
 - Total market movement
-

About Sensex and Nifty

- **Sensex (Sensitive Index)** is the benchmark index of the Bombay Stock Exchange (BSE). It consists of 30 well-established and financially sound companies across major sectors of the Indian economy.
- **Nifty 50** is the benchmark index of the National Stock Exchange (NSE). It includes 50 major companies representing various sectors.

Both indices act as indicators of overall market performance and economic conditions in India.

Chart.js Library

Chart.js is a powerful and lightweight JavaScript library used for creating interactive charts.

Features of Chart.js:

- Easy integration using CDN
 - Supports multiple chart types
 - Responsive and mobile-friendly
 - Animated transitions
 - Customizable design options
 - Real-time data updates
-

Chart Types Used in the Dashboard

In the Sensex and Nifty Dashboard, the following chart types are used:

- **Line Chart** – To show stock price trends over time (daily/weekly/monthly).
 - **Bar Chart** – To compare closing prices of Sensex and Nifty.
 - **Area Chart** – To visualize overall market growth.
 - **Doughnut/Pie Chart** – To represent percentage change or sector distribution.
-

Working of the Dashboard

The dashboard works as follows:

1. Stock data (Sensex and Nifty values) is stored inside the JavaScript file or fetched from a dataset/API.

2. When the webpage loads, charts are automatically generated using Chart.js.
3. Summary statistics such as average price, highest value, and lowest value are calculated dynamically using JavaScript.
4. When the user clicks:
 - **“Weekly View”** → Displays weekly stock data.
 - **“Monthly View”** → Displays monthly stock data.
5. Charts update dynamically without refreshing the page.

This interactive approach allows users to easily switch between different time periods and analyze stock market performance efficiently.

Importance of Stock Market Dashboards

- Helps investors track market performance in real time
- Makes complex financial data easy to understand
- Supports data-driven investment decisions
- Identifies trends and patterns quickly
- Improves financial analysis skills

Code:

HTML-

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>SENSEX & NIFTY Dashboard</title>
  <link rel="preconnect" href="https://fonts.googleapis.com">
  <link
href="https://fonts.googleapis.com/css2?family=Space+Mono:wght@400;700&family=
DM+Sans:wght@300;400;500;600;700&display=swap" rel="stylesheet">
  <script src="https://cdn.jsdelivr.net/npm/chart.js"></script>
  <link rel="stylesheet" href="styles.css">
</head>
```

```
<body>

<div class="bg-grid"></div>

<div class="container">

  <header class="header">
    <div class="header-badge">LIVE MARKET</div>
    <h1 class="header-title">Indian Market<br></h1>
    <p class="header-subtitle">SENSEX & NIFTY 50 — Intraday
Performance</p>
    <div class="header-meta">
    </div>
  </header>

  <div class="stats-grid">

    <div class="stat-card" id="sensexCard">
      <div class="stat-label">BSE SENSEX</div>
      <div class="stat-value" id="sensexValue">—</div>
      <div class="stat-change" id="sensexChange">—</div>
      <div class="stat-range">
        <span>High: <strong id="sensexHigh">—</strong></span>
        <span>Low: <strong id="sensexLow">—</strong></span>
      </div>
    </div>

    <div class="stat-card" id="niftyCard">
      <div class="stat-label">NIFTY 50</div>
      <div class="stat-value" id="niftyValue">—</div>
      <div class="stat-change" id="niftyChange">—</div>
      <div class="stat-range">
        <span>High: <strong id="niftyHigh">—</strong></span>
        <span>Low: <strong id="niftyLow">—</strong></span>
      </div>
    </div>

  </div>

  <div class="charts-grid">
    <div class="chart-card">
      <div class="chart-header">
        <h2>SENSEX Trend</h2>
      </div>
    </div>
  </div>
</div>
```

```

        <div class="chart-pill sensex-pill">BSE</div>
    </div>
    <div class="chart-container">
        <canvas id="sensexChart"></canvas>
    </div>
</div>

<div class="chart-card">
    <div class="chart-header">
        <h2>NIFTY 50 Trend</h2>
        <div class="chart-pill nifty-pill">NSE</div>
    </div>
    <div class="chart-container">
        <canvas id="niftyChart"></canvas>
    </div>
</div>
</div>

<script src="data.js"></script>
<script src="app.js"></script>
</body>
</html>

```

CSS-

```
*, *::before, *::after { margin: 0; padding: 0; box-sizing: border-box; }
```

```

:root {
    --bg:      #080c14;
    --surface: #0e1420;
    --surface-2: #151d2e;
    --border:  #1e293b;
    --border-hl: #2d4263;

    --text:    #dce8f5;
    --muted:   #5a7a9a;

    --blue:    #3b82f6;
    --blue-glow: rgba(59,130,246,0.25);
    --violet:   #8b5cf6;
    --violet-glow: rgba(139,92,246,0.25);

```

```
--green:    #22d07a;
--red:      #f04f5e;

--font-mono: 'Space Mono', monospace;
--font-body: 'DM Sans', sans-serif;

--radius:   14px;
--radius-sm: 8px;
}

body {
  font-family: var(--font-body);
  background: var(--bg);
  color: var(--text);
  min-height: 100vh;
  padding: 32px 20px 60px;
  overflow-x: hidden;
}

.bg-grid {
  position: fixed;
  inset: 0;
  background-image:
    linear-gradient(rgba(59,130,246,0.03) 1px, transparent 1px),
    linear-gradient(90deg, rgba(59,130,246,0.03) 1px, transparent 1px);
  background-size: 40px 40px;
  pointer-events: none;
  z-index: 0;
}

.container {
  max-width: 1360px;
  margin: 0 auto;
  position: relative;
  z-index: 1;
}

.header {
  text-align: center;
  margin-bottom: 48px;
  padding-top: 24px;
}

.header-badge {
```

```

display: inline-block;
font-family: var(--font-mono);
font-size: 0.65rem;
letter-spacing: 3px;
color: var(--blue);
border: 1px solid rgba(59,130,246,0.4);
padding: 4px 14px;
border-radius: 99px;
margin-bottom: 20px;
background: rgba(59,130,246,0.07);
animation: pulse-border 3s ease-in-out infinite;
}

@keyframes pulse-border {
  0%, 100% { border-color: rgba(59,130,246,0.3); }
  50% { border-color: rgba(59,130,246,0.9); box-shadow: 0 0 12px
  rgba(59,130,246,0.3); }
}

.header-title {
  font-family: var(--font-mono);
  font-size: clamp(2.8rem, 6vw, 5rem);
  font-weight: 700;
  line-height: 1.05;
  letter-spacing: -2px;
  color: var(--text);
  margin-bottom: 14px;
}

.header-title .accent {
  background: linear-gradient(135deg, var(--blue) 0%, var(--violet) 100%);
  -webkit-background-clip: text;
  -webkit-text-fill-color: transparent;
  background-clip: text;
}

.header-subtitle {
  color: var(--muted);
  font-size: 1rem;
  letter-spacing: 0.5px;
  margin-bottom: 16px;
}

.header-meta {

```

```
display: flex;
align-items: center;
justify-content: center;
gap: 8px;
font-size: 0.78rem;
color: var(--muted);
font-family: var(--font-mono);
}
```

```
.dot {
width: 8px;
height: 8px;
border-radius: 50%;
display: inline-block;
}
```

```
.positive-dot { background: var(--green); box-shadow: 0 0 8px var(--green); animation:
blink 2s step-end infinite; }
```

```
@keyframes blink {
0%, 100% { opacity: 1; }
50%     { opacity: 0.2; }
}
```

```
.stats-grid {
display: grid;
grid-template-columns: repeat(2, 1fr);
gap: 16px;
margin-bottom: 20px;
}
```

```
@media (max-width: 620px) {
.stats-grid { grid-template-columns: 1fr; }
}
```

```
.stat-card {
background: var(--surface);
border: 1px solid var(--border);
border-radius: var(--radius);
padding: 28px 24px;
transition: transform 0.25s ease, border-color 0.25s ease, box-shadow 0.25s ease;
position: relative;
overflow: hidden;
}
```

```
.stat-card::before {  
  content: "";  
  position: absolute;  
  top: 0; left: 0; right: 0;  
  height: 2px;  
  background: linear-gradient(90deg, var(--blue), var(--violet));  
  opacity: 0;  
  transition: opacity 0.25s;  
}
```

```
.stat-card:hover {  
  transform: translateY(-4px);  
  border-color: var(--border-hl);  
  box-shadow: 0 12px 32px rgba(0,0,0,0.4);  
}
```

```
.stat-card:hover::before { opacity: 1; }
```

```
.stat-label {  
  font-size: 0.72rem;  
  font-family: var(--font-mono);  
  letter-spacing: 2px;  
  text-transform: uppercase;  
  color: var(--muted);  
  margin-bottom: 12px;  
}
```

```
.stat-value {  
  font-family: var(--font-mono);  
  font-size: 2rem;  
  font-weight: 700;  
  color: var(--text);  
  margin-bottom: 8px;  
  letter-spacing: -1px;  
}
```

```
.stat-change {  
  font-size: 1rem;  
  font-weight: 600;  
  margin-bottom: 16px;  
  font-family: var(--font-mono);  
}
```

```
.positive { color: var(--green); }
.negative { color: var(--red); }

.stat-range {
  display: flex;
  justify-content: space-between;
  font-size: 0.72rem;
  color: var(--muted);
}

.stat-range strong { color: var(--text); font-weight: 600; }

.stat-card--mini {
  display: flex;
  flex-direction: column;
}

.sentiment-bar-wrap { margin: 8px 0 4px; }

.sentiment-bar {
  height: 8px;
  background: rgba(240,79,94,0.25);
  border-radius: 99px;
  overflow: hidden;
  margin-bottom: 6px;
}

.sentiment-fill {
  height: 100%;
  border-radius: 99px;
  background: linear-gradient(90deg, var(--red), var(--green));
  transition: width 1s ease;
}

.sentiment-labels {
  display: flex;
  justify-content: space-between;
  font-size: 0.7rem;
  font-family: var(--font-mono);
  letter-spacing: 1px;
}

.adv-decline {
  display: flex;
```

```
    align-items: baseline;
    gap: 8px;
    font-family: var(--font-mono);
    font-size: 1.6rem;
    font-weight: 700;
    margin-top: 4px;
}

.divider { color: var(--muted); font-size: 1.2rem; }

.mini-stats { display: flex; flex-direction: column; gap: 10px; margin-top: 4px; }

.mini-stat {
    display: flex;
    justify-content: space-between;
    font-size: 0.82rem;
    color: var(--muted);
    border-bottom: 1px solid var(--border);
    padding-bottom: 6px;
}

.mini-stat strong { color: var(--text); font-family: var(--font-mono); font-size: 0.82rem; }

.charts-grid {
    display: grid;
    grid-template-columns: repeat(2, 1fr);
    gap: 16px;
    margin-bottom: 20px;
}

@media (max-width: 900px) {
    .charts-grid { grid-template-columns: 1fr; }
}

.chart-card {
    background: var(--surface);
    border: 1px solid var(--border);
    border-radius: var(--radius);
    padding: 28px 24px;
}

.chart-header {
    display: flex;
    align-items: center;
```

```
    justify-content: space-between;
    margin-bottom: 20px;
}
```

```
.chart-header h2 {
    font-size: 1.05rem;
    font-weight: 600;
    letter-spacing: -0.3px;
    color: var(--text);
}
```

```
.chart-pill {
    font-family: var(--font-mono);
    font-size: 0.6rem;
    letter-spacing: 2px;
    padding: 3px 10px;
    border-radius: 99px;
    font-weight: 700;
}
```

```
.sensex-pill { background: var(--blue-glow); color: var(--blue); border: 1px solid
rgba(59,130,246,0.4); }
.nifty-pill { background: var(--violet-glow); color: var(--violet); border: 1px solid
rgba(139,92,246,0.4); }
```

```
.chart-container {
    position: relative;
    height: 300px;
}
```

```
.table-card {
    background: var(--surface);
    border: 1px solid var(--border);
    border-radius: var(--radius);
    padding: 28px 24px;
    margin-bottom: 20px;
}
```

```
.table-title {
    font-size: 1.05rem;
    font-weight: 600;
    color: var(--text);
    margin-bottom: 20px;
    letter-spacing: -0.3px;
}
```

```
}
```

```
.table-wrap { overflow-x: auto; }
```

```
.movers-table {  
  width: 100%;  
  border-collapse: collapse;  
  font-size: 0.85rem;  
}
```

```
.movers-table th {  
  text-align: left;  
  font-family: var(--font-mono);  
  font-size: 0.65rem;  
  letter-spacing: 1.5px;  
  text-transform: uppercase;  
  color: var(--muted);  
  padding: 10px 14px;  
  border-bottom: 1px solid var(--border);  
}
```

```
.movers-table td {  
  padding: 12px 14px;  
  border-bottom: 1px solid rgba(30,41,59,0.5);  
  vertical-align: middle;  
}
```

```
.movers-table tbody tr:hover { background: var(--surface-2); }
```

```
.stock-name { font-weight: 600; color: var(--text); font-size: 0.88rem; }  
.stock-sector { font-size: 0.7rem; color: var(--muted); font-family: var(--font-mono); }
```

```
.change-badge {  
  display: inline-block;  
  padding: 3px 8px;  
  border-radius: var(--radius-sm);  
  font-family: var(--font-mono);  
  font-size: 0.75rem;  
  font-weight: 700;  
}
```

```
.badge-pos { background: rgba(34,208,122,0.12); color: var(--green); }  
.badge-neg { background: rgba(240,79,94,0.12); color: var(--red); }
```

```
.footer {
  text-align: center;
  color: var(--muted);
  font-size: 0.72rem;
  font-family: var(--font-mono);
  letter-spacing: 0.5px;
  margin-top: 40px;
}
```

Java Script-

```
(function () {
  "use strict";

  function fmt(num) {
    return num.toLocaleString("en-IN", { minimumFractionDigits: 2,
    maximumFractionDigits: 2 });
  }

  function calcChange(prices, prevClose) {
    const current = prices[prices.length - 1];
    const change = current - prevClose;
    const pct = (change / prevClose) * 100;
    return { current, change, pct };
  }

  function arrHigh(arr) { return Math.max(...arr); }
  function arrLow(arr) { return Math.min(...arr); }

  function setEl(id, html) {
    const el = document.getElementById(id);
    if (el) el.innerHTML = html;
  }

  function addClass(id, cls) {
    const el = document.getElementById(id);
    if (el) el.classList.add(cls);
  }

  function renderStats() {
    const s = MARKET_DATA.sensex;
    const n = MARKET_DATA.nifty;
    const b = MARKET_DATA.breadth;
```

```

const sensex = calcChange(s.prices, s.prevClose);
const nifty = calcChange(n.prices, n.prevClose);

setEl("sensexValue", `₹${fmt(sensex.current)}`);
const sDir = sensex.change >= 0 ? "▲" : "▼";
const sClass = sensex.change >= 0 ? "positive" : "negative";
setEl("sensexChange",
  `

```

```

        tooltip: {
            backgroundColor: "#0e1420",
            titleColor: "#5a7a9a",
            bodyColor: "#dce8f5",
            borderColor: "#2d4263",
            borderWidth: 1,
            padding: 12,
            callbacks: {
                label: ctx => ` ₹${ctx.parsed.y.toLocaleString("en-IN", {
minimumFractionDigits: 2 })}`
            }
        },
        scales: {
            x: {
                grid: { color: "rgba(30,41,59,0.6)", drawBorder: false },
                ticks: { color: "#5a7a9a", font: { family: "'Space Mono', monospace", size: 10
}}}
        },
        y: {
            beginAtZero: false,
            position: "right",
            grid: { color: "rgba(30,41,59,0.6)", drawBorder: false },
            ticks: {
                color: "#5a7a9a",
                font: { family: "'Space Mono', monospace", size: 10 },
                callback: v => "₹" + v.toLocaleString("en-IN")
            }
        }
    },
    elements: {
        point: { radius: 0, hoverRadius: 5, hoverBorderWidth: 2 }
    }
};

```

```

function renderCharts() {
    const labels = MARKET_DATA.timeLabels;

    const sensexCtx = document.getElementById("sensexChart").getContext("2d");
    new Chart(sensexCtx, {
        type: "line",
        data: {
            labels,
            datasets: [{

```

```

        label: "SENSEX",
        data: MARKET_DATA.sensex.prices,
        borderColor: "#3b82f6",
        backgroundColor: makeGradient(sensexCtx, "59,130,246"),
        tension: 0.35,
        fill: true,
        borderWidth: 2
    ]
},
options: commonOptions
});

// NIFTY chart
const niftyCtx = document.getElementById("niftyChart").getContext("2d");
new Chart(niftyCtx, {
    type: "line",
    data: {
        labels,
        datasets: [{
            label: "NIFTY 50",
            data: MARKET_DATA.nifty.prices,
            borderColor: "#8b5cf6",
            backgroundColor: makeGradient(niftyCtx, "139,92,246"),
            tension: 0.35,
            fill: true,
            borderWidth: 2
        }]
    },
    options: commonOptions
});
}

function init() {
    renderStats();
    renderCharts();
}

if (document.readyState === "loading") {
    document.addEventListener("DOMContentLoaded", init);
} else {
    init();
}

})();

```

Dataset:

```
const MARKET_DATA = {
```

```
  timeLabels: [  
    "9:15","9:30","9:45","10:00","10:15","10:30","10:45",  
    "11:00","11:15","11:30","11:45","12:00","12:15","12:30",  
    "12:45","13:00","13:15","13:30","13:45","14:00","14:15",  
    "14:30","14:45","15:00","15:15","15:30"  
  ],
```

```
  sensex: {  
    prices: [  
      79521.40, 79618.75, 79542.10, 79710.55, 79834.20, 79902.60, 79988.30,  
      80105.45, 80240.10, 80318.85, 80197.50, 80070.20, 80143.60, 80062.30,  
      80011.75, 80090.40, 80204.15, 80289.70, 80354.20, 80412.85, 80476.50,  
      80389.10, 80302.40, 80440.65, 80521.90, 80485.30  
    ],  
    high52w: 85978.25,  
    low52w: 71896.50,  
    prevClose: 79387.20  
  },
```

```
  nifty: {  
    prices: [  
      24108.30, 24154.75, 24122.40, 24198.60, 24261.85, 24295.40, 24338.10,  
      24402.55, 24475.20, 24511.80, 24448.30, 24381.60, 24420.90, 24378.45,  
      24349.70, 24398.25, 24452.10, 24488.60, 24531.40, 24568.85, 24601.30,  
      24557.90, 24504.20, 24572.45, 24619.80, 24598.10  
    ],  
    high52w: 26277.35,  
    low52w: 21964.60,  
    prevClose: 24055.60  
  },
```

```
  breadth: {  
    advancers: 1842,  
    decliners: 897,  
    unchanged: 124,  
    volume: 4218.6,  
    turnover: 87432.5  
  },
```

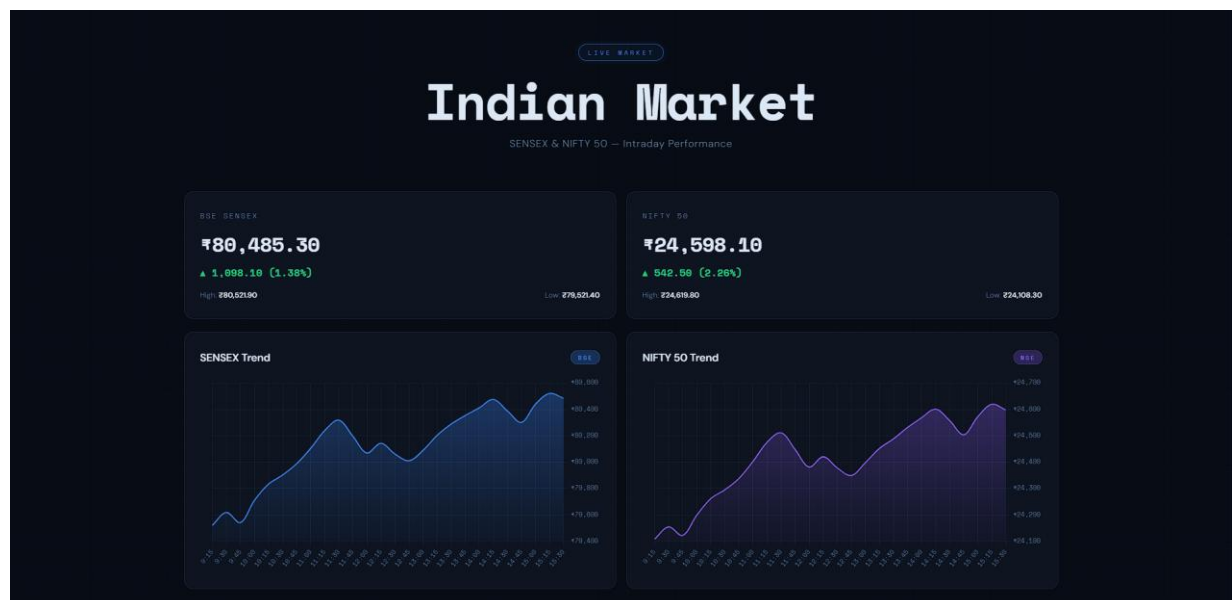
```
  topMovers: [  
    "9:15","9:30","9:45","10:00","10:15","10:30","10:45",  
    "11:00","11:15","11:30","11:45","12:00","12:15","12:30",  
    "12:45","13:00","13:15","13:30","13:45","14:00","14:15",  
    "14:30","14:45","15:00","15:15","15:30"
```

```

    { name: "RELIANCE", sector: "Energy",    price: 2847.55, change: +62.30, pct:
+2.24, volume: "28.4L" },
    { name: "HDFC BANK", sector: "Banking",    price: 1724.90, change:
+34.15, pct: +2.02, volume: "19.1L" },
    { name: "INFOSYS", sector: "IT",          price: 1638.45, change: -18.60, pct: -
1.12, volume: "14.7L" },
    { name: "TCS", sector: "IT",          price: 3892.10, change: -42.50, pct: -1.08,
volume: "9.8L" },
    { name: "ICICI BANK",sector: "Banking",    price: 1245.30, change: +21.80, pct:
+1.78, volume: "22.3L" },
    { name: "BAJFINANCE",sector: "NBFC",      price: 7214.60, change: +189.40,
pct: +2.70, volume: "7.2L" },
    { name: "WIPRO", sector: "IT",          price: 468.75, change: -6.45, pct: -
1.36, volume: "31.5L" },
    { name: "TITAN", sector: "Consumer",    price: 3421.50, change: +98.25, pct:
+2.96, volume: "5.9L" },
    { name: "LT", sector: "Infra",          price: 3687.40, change: +54.70, pct: +1.51,
volume: "6.1L" },
    { name: "NESTLEIND", sector: "FMCG",      price: 2318.85, change: -
29.35, pct: -1.25, volume: "3.4L" }
  ]
};

```

OUTPUT:



Conclusion:

The Sensex and Nifty Interactive Data Visualization Dashboard shows how stock market data can be clearly understood through graphical representation. Using HTML, CSS, JavaScript, and Chart.js, an interactive and user-friendly dashboard was created to display market trends and index movements.

By providing weekly and monthly views along with summary statistics, the dashboard helps users easily analyze stock performance. Overall, the project demonstrates how data visualization makes financial data simpler, more accessible, and useful for better decision-making.